

The **optmodel** package

typesetting optimization models

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1 Introduction

optmodel provides one environment, `optmodel`, that lays out a mathematical optimization model: a keyword (`\min, s.t.`), the left-hand sides, relations and right-hand sides of the constraints aligned in columns, an optional quantifier column (`\forall \tau`), optional per-line numbers or tags, and an optional tag for the whole model. The block is centred (or indented) as a unit and never exceeds the line width.

All public names carry the prefix **opt**, so the package is safe to load next to other packages. It needs a L^AT_EX kernel from 2020-10-01 or later and loads `amsmath`, `mathtools` and `array`.

2 Syntax

```
\usepackage{optmodel}  
...  
\begin{optmodel}[<model tag>][<label key>]  
  <keyword> & <lhs> & <rel> & <rhs> & <quantifier> & <line tag> \\  
  ...  
\end{optmodel}
```

Every row has up to six columns, separated by `&`: (1) keyword, (2) left-hand side (right-aligned), (3) relation, (4) right-hand side, (5) quantifier, (6) line number or tag (right-aligned at the margin). Columns 2–5 are in display math mode; column 1 is too, so use `\text{s.t.}` for words. Missing trailing cells may be omitted. Rows are separated by `\\`; the space between rows is `\optrowsep` unless you give the usual optional argument, `\\[2pt]`.

Both optional arguments of the environment are optional:

Call	Effect
<code>\begin{optmodel}</code>	no model tag
<code>\begin{optmodel}[PP2]</code>	model tag (PP2) at the right margin
<code>\begin{optmodel}[]</code>	auto-numbered model tag, uses the equation counter
<code>\begin{optmodel}[...][key]</code>	additionally <code>\label{key}</code> , so <code>\ref{key}</code> yields the tag

3 Modes

All examples use the same small model unless noted.

3.1 No label

```
\begin{optmodel}
  \min      & \optobjective{c^{\top} x} \\
  \text{s.t.} & A x \leq b \\
            & x \geq 0 \\
\end{optmodel}
```

$$\begin{array}{ll}
 \min & c^\top x \\
 \text{s.t.} & Ax \leq b \\
 & x \geq 0
 \end{array}$$

3.2 Custom label

```
\begin{optmodel}[LP]
  ...
\end{optmodel}
```

$$\begin{array}{ll}
 \min & c^\top x \\
 \text{s.t.} & Ax \leq b \\
 & x \geq 0
 \end{array} \tag{LP}$$

3.3 Automatic label and a label key

An empty first argument takes the next equation number; the second argument is a `\label` key.

```
\begin{optmodel}[] [m:lp]
...
\end{optmodel}
Model~\ref{m:lp} is a linear program.
```

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0 \end{array} \quad (1)$$

Model 1 is a linear program.

3.4 Per-line numbers: `\optnum`

Put `\optnum` in column 6 to number a line with the next equation number. Labels work as usual.

```
\begin{optmodel}
\min & & & & & \& \optobjective{c^{\top} x} \& \& \\
\text{s.t.} & & A x & \& \leq & b & \& \& \optnum\label{c:cap} \& \& \\
& & & & x & \& \geq & 0 & \& \& \optnum \\
\end{optmodel}
Constraint~\ref{c:cap} is the capacity constraint.
```

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0 \end{array} \quad \begin{array}{l} (2) \\ (3) \end{array}$$

Constraint 2 is the capacity constraint.

3.5 Per-line tags: `\opttag`, with a model tag

`\opttag{<text>}` prints a fixed tag; `\label` then refers to that text. A row consisting of a single `\optwholerow` spans columns 2–4 and is left aligned; it is meant for domains.

```
\begin{optmodel}[PP2] [m:pp2]
\min & & & & & \& \optobjective{\sum_t c_t x_t} \& \& \\
\text{s.t.} & & \sum_t A^{\mathrm{I}}_t x_t & \& \leq & b^{\mathrm{I}} & \& \& \opttag{I}\label{c:I} \& \& \\
& & A^{\mathrm{II}}_t x_t + B_t y_t & \& \leq & b^{\mathrm{II}}_t, \\
& & \forall t & \& \opttag{II}\label{c:II} \& \& \\
& & \optwholerow{x_t \in \mathbf{X}_t, y_t \in \mathbf{Y}_t} & \& \& \& \\
\end{optmodel}
Model~\ref{m:pp2} has linking constraints~\ref{c:I} and~\ref{c:II}.
```

$$\begin{aligned}
& \min && \sum_t c_t x_t \\
& \text{s.t.} && \sum_t A_t^I x_t \leq b^I && \text{(I)} \\
& && A_t^{\text{II}} x_t + B_t y_t \leq b_t^{\text{II}}, \quad \forall t && \text{(II)} \\
& && x_t \in \mathbf{X}_t, \quad y_t \in \mathbf{Y}_t
\end{aligned} \tag{PP2}$$

Model PP2 has linking constraints I and II.

3.6 Centred versus indented

By default the block is centred in the line (switch `\optcentertrue`). After `\optcenterfalse` it is instead indented by `\optindent` from the left margin. The indent is reduced when the model would otherwise run into the tags. Both switches and lengths are ordinary $\text{\TeX}$ assignments, so keep them in a group.

```

\begin{group}
  \optcenterfalse \setlength{\optindent}{3em}
  \begin{optmodel}[LP] ... \end{optmodel}
\end{group}

```

$$\begin{aligned}
& \min && c^\top x \\
& \text{s.t.} && Ax \leq b \\
& && x \geq 0
\end{aligned} \tag{LP}$$

3.7 Long objective

`\optobjective` spans columns 2–4 but adds nothing to their widths: the constraints alone determine the layout, and a long objective starts at the left edge of the left-hand-side column and simply extends to the right of the constraints. The line numbers below belong to the constraints only.

```

\begin{optmodel}
  \min && & \optobjective{\sum_{t=1}^T \sum_{k=1}^K}
  \bigl(c_{\text{tk}}x_{\text{tk}}+f_{\text{tk}}y_{\text{tk}}+h_{\text{tk}}s_{\text{tk}}\bigr)
  + \sum_{t=1}^T p_t u_t \\
  \text{s.t.} & s_{t-1,k}+x_{\text{tk}}-s_{\text{tk}} & = & d_{\text{tk}}, \\
  & \forall t,k & \text{\optnum} & \\
  & x_{\text{tk}} & \leq & M y_{\text{tk}}, \quad \forall t,k & \text{\optnum} \\
\end{optmodel}

```

$$\begin{aligned}
\min \quad & \sum_{t=1}^T \sum_{k=1}^K (c_{tk}x_{tk} + f_{tk}y_{tk} + h_{tk}s_{tk}) + \sum_{t=1}^T p_t u_t \\
\text{s.t.} \quad & s_{t-1,k} + x_{tk} - s_{tk} = d_{tk}, \quad \forall t, k \\
& x_{tk} \leq M y_{tk}, \quad \forall t, k
\end{aligned}
\tag{4}$$

$$\tag{5}$$

4 Parameters

The spacing parameters are ordinary lengths and the centring is a switch. They may be changed at any point and apply to the models that follow in the same group.

Name	Default	Meaning
<code>\optkeysep</code>	1em	gap between keyword and left-hand side
<code>\optquantsep</code>	1em	gap between right-hand side and quantifier
<code>\optnumsep</code>	2em	minimum gap before the line numbers
<code>\opttagsep</code>	1.5em	gap between line numbers and the model tag
<code>\optrowsep</code>	4pt	extra space between rows
<code>\optindent</code>	2em	left indent when <code>\optcenterfalse</code> is in effect

Command	Meaning
<code>\optobjective{...}</code>	objective row, starts at the left-hand-side column, adds no width
<code>\optwholerow{...}</code>	free row across columns 2–4, left aligned, adds width
<code>\optnum</code>	auto-numbered line tag (column 6)
<code>\opttag{...}</code>	fixed-text line tag (column 6)
<code>\optcentertrue</code>	centre the block (default)
<code>\optcenterfalse</code>	indent the block by <code>\optindent</code>

5 Limitations

- The body is read as an argument and typeset twice (once to measure it). Verbatim material (`\verb`, `listings`) is not allowed inside, and commands with side effects other than `\label`, `\optnum` and `\opttag` run twice.

- A model is a single unbreakable block; it will not be split across pages.
- An objective is not line-broken. One that is wider than the space to the right of its column runs past the margin or into the model tag, and $\text{\TeX}$ does *not* report this as an overfull box. Check long objectives by eye and split them by hand.
- The layout has the fixed six columns described above.
- With `hyperref` the package compiles cleanly and `\ref` prints the right text, but a `\label` after `\opttag` links to the enclosing section, not to the line. Model tags and `\optnum` lines link correctly.
- The environment builds on `equation*`; do not nest it in other display environments.